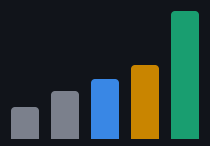
HIT SONG PREDICTION · TECHNION · PROJECT IN AI

What really predicts a hit?

Artist history, era context and shortcut features — tested on four datasets, with every gain checked for chance and for artefacts.

Lior Shalon & Nathanael Uzzan

Technion – Israel Institute of Technology · Project in Artificial Intelligence



THE PROBLEM

Hit-prediction papers report 86 %, 88 %, 89 % — and none of these numbers can be compared

≠

Different datasets

Each study builds its own hits and non-hits. An 88 % on one set says nothing about another.

1×

One split, no test

Gains are measured on a single train/test split, usually without a significance test.



Shortcuts

A feature can separate hits from non-hits because of **how the negatives were chosen**, not because of anything that makes a song succeed.

Our question: which gains are **real** — and how big are they once all three problems are controlled?

DATA

Four datasets — the definition of "hit" changes every time

DATASET	SONGS	HITS	HIT LABEL	HOW NON-HITS WERE CHOSEN
Spotify Hit Predictor (Kaggle, 1960–2019)	40,560	49 %	Weekly Hot 100 in that decade	Rules: artist *and* genre absent from the hit list
audio subset (30 s Deezer clips)	3,562	52 %	same	same
ISMIR 2019 (Zangerle et al., peer-reviewed)	11,662	50 %	Hot 100 match	Random Million Song Dataset tracks
SpotGenTrack	101,909	19 %	Spotify popularity ≥ 55	Everything else
~~Billboard year-end 2000–2023~~	3,397		DROPPED 2,911 rows have no audio features	

Paper · Table I

Two cheap, non-audio signals added to the 13 Spotify audio features

ERA

Decade / release year

A hit in the 1960s does not sound like a hit in the 2010s. The era lets the model apply **era-specific rules**.

One-hot decade on the Hit Predictor; year or decade as a number elsewhere.

ARTIST HISTORY

Earlier Hot 100 songs

```
h(s) = log(1 + #hits by the same artist
strictly before the song's period)
```

Strictly earlier periods only → **no information from the song's own period or the future.**

Artist names normalised (accents, "feat.", "&" removed); 52 % of benchmark songs match the chart history.

METHOD · HOW EVERY GAIN IS TESTED

Two estimates for every comparison — and a test that is not fooled by overlap

1 · Canonical split

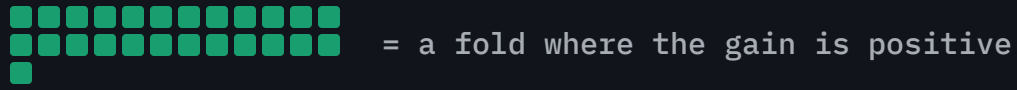
The split of our earlier experiments (seed 42; 20,000 train / 2,000 test). Two models compared song by song with **McNemar's test**.



Reproduces the old numbers exactly: .773 → .822.

2 · Repeated cross-validation

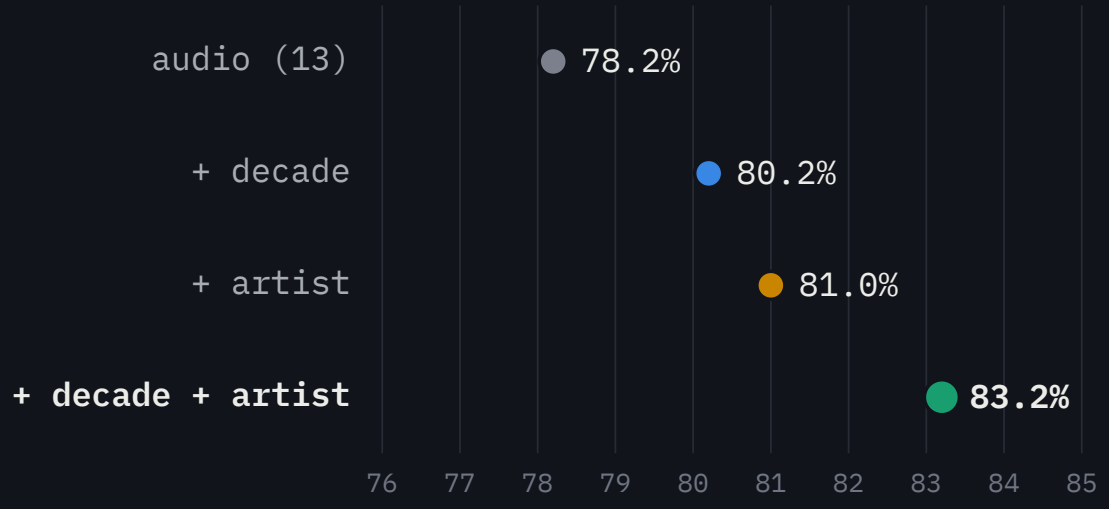
5 folds × 5 repeats = **25 test sets**. Mean, 95 % confidence interval, and the **Nadeau–Bengio corrected t-test** — the naive t-test is over-confident because the folds overlap.



Scaling, PCA and title models are fitted inside each training fold only.

RESULT 1 · SPOTIFY HIT PREDICTOR, 40,560 SONGS

Decade and artist add +5.0 points, and the gains stack



Random Forest · 5×5 CV mean accuracy · paper Table II

83.2%

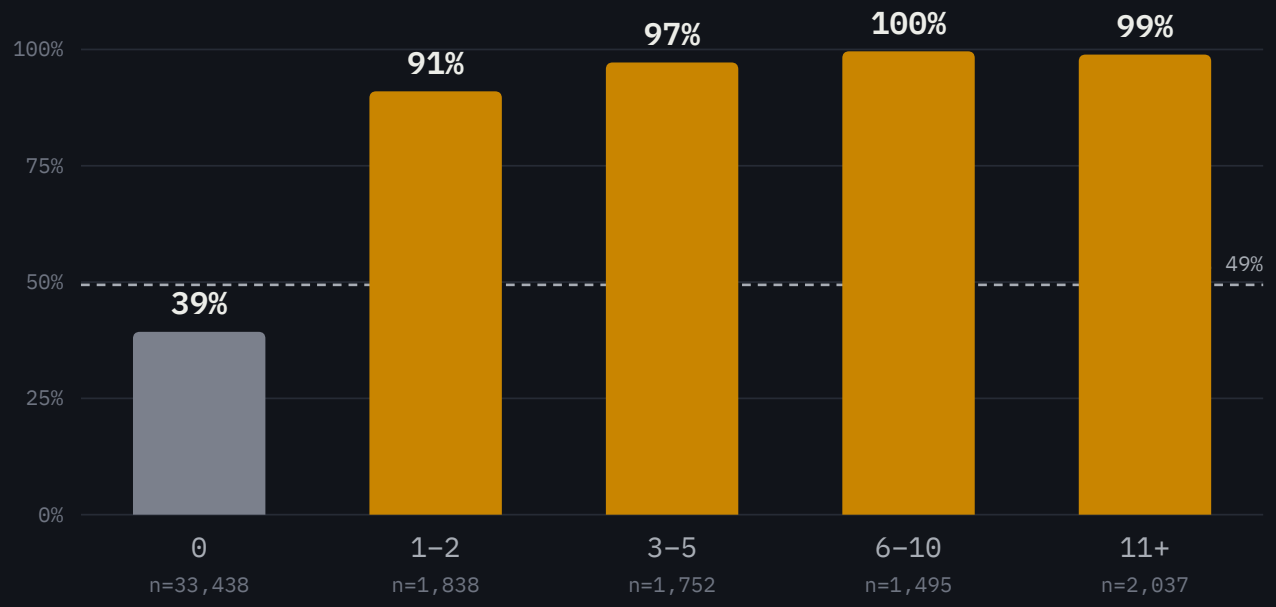
from 78.2 % with audio alone · ROC-AUC .859 → .918

25 / 25 folds

$p = 7 \cdot 10^{-17}$

Canonical split: .773 → .796 → .800 → .822. The two signals alone reach only 67.0 %: they complement the audio, they do not replace it.

One earlier hit changes everything: 39 % → 91 %



Share of songs that are hits, by the artist's Hot 100 songs in earlier decades · paper Fig. 1

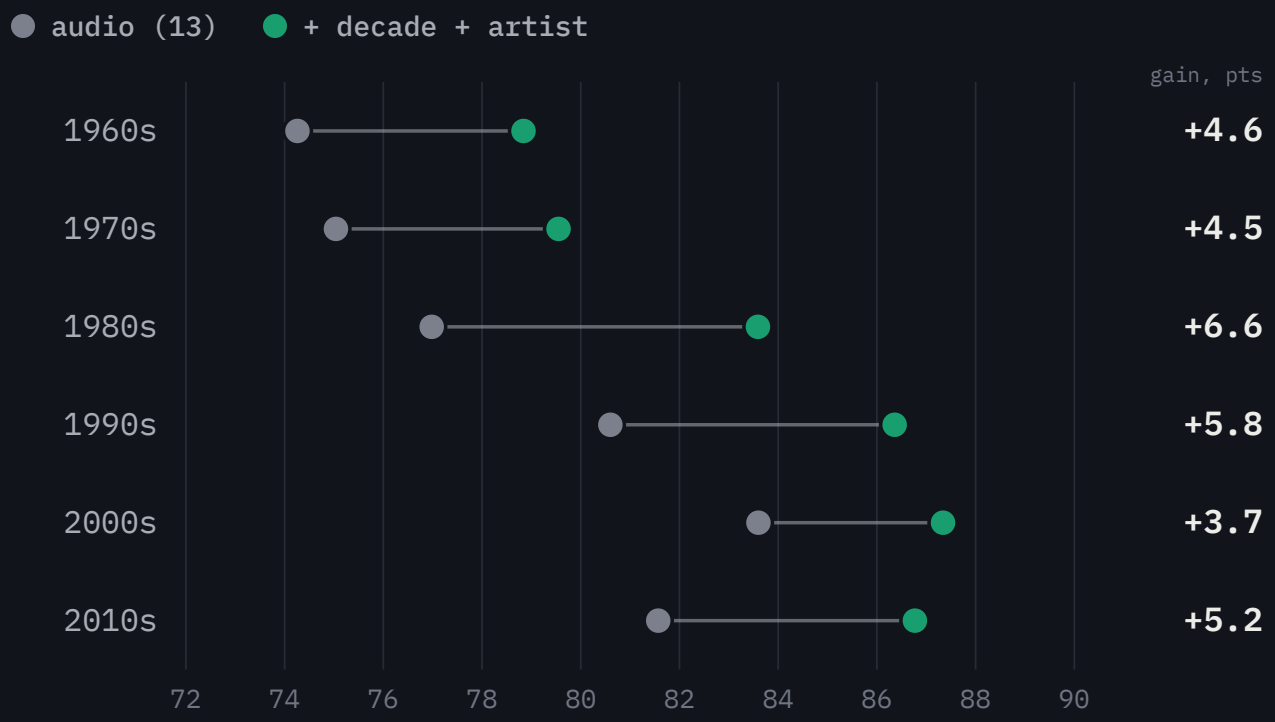
A strong, very non-linear relation: the first hits matter most, then it plateaus near 100 %.

MEAN EARLIER HITS

hits 3.21 · non-hits 0.037

Keep that 0.037 in mind — it comes back on slide 12.

The gain holds in all six decades, from +3.7 to +6.6 points



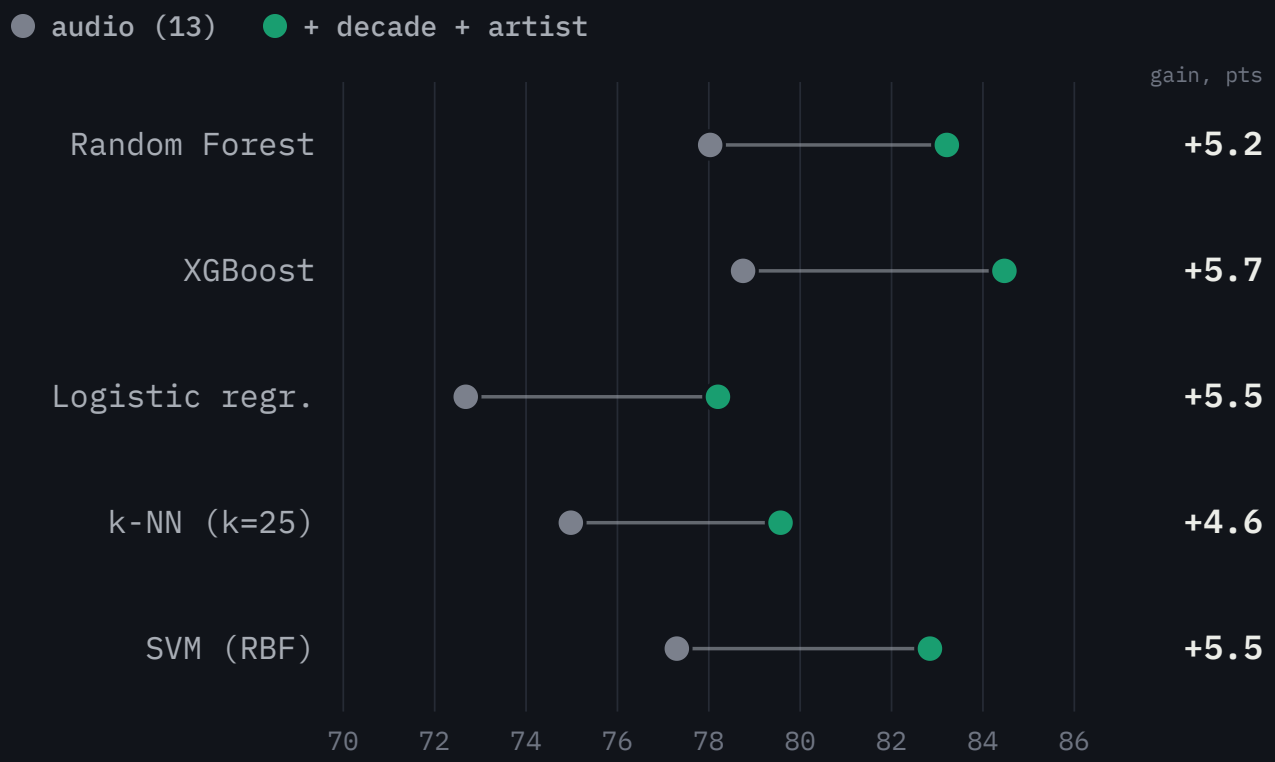
5×5 CV accuracy per decade · paper Table III

Each decade file is ~50 % hits, so the decade alone says nothing.

Its gain can only come from **interactions with the audio** — the model learns what a hit sounded like in that era.

ROBUSTNESS · EVERY CLASSIFIER

Five very different models, the same +5 points: it is the features, not the model



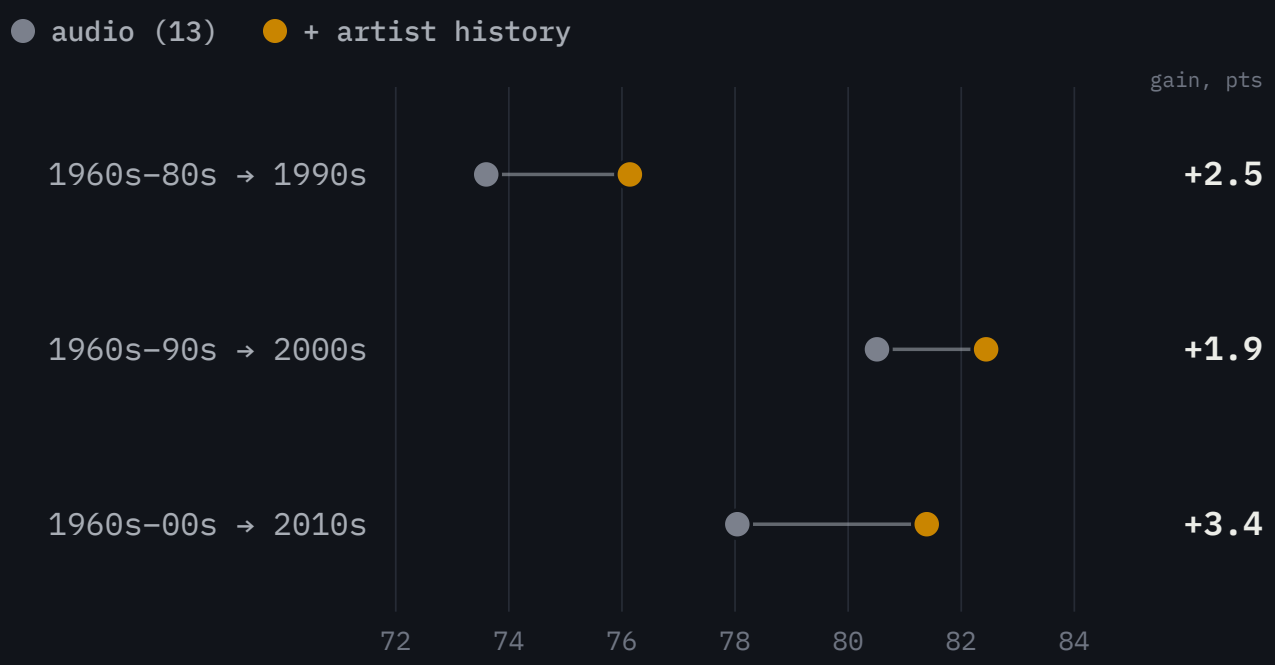
5-fold CV, corrected t-test, all 5 folds improve · paper Table IV

84.5%

XGBoost is the strongest on average — but on the canonical split it ties with the Random Forest (.820 vs .822, p = .87).

TreeSHAP ranks **artist history #1**, the six decade indicators #2 and instrumentalness #3.

Train on the past, predict a decade the model has never seen

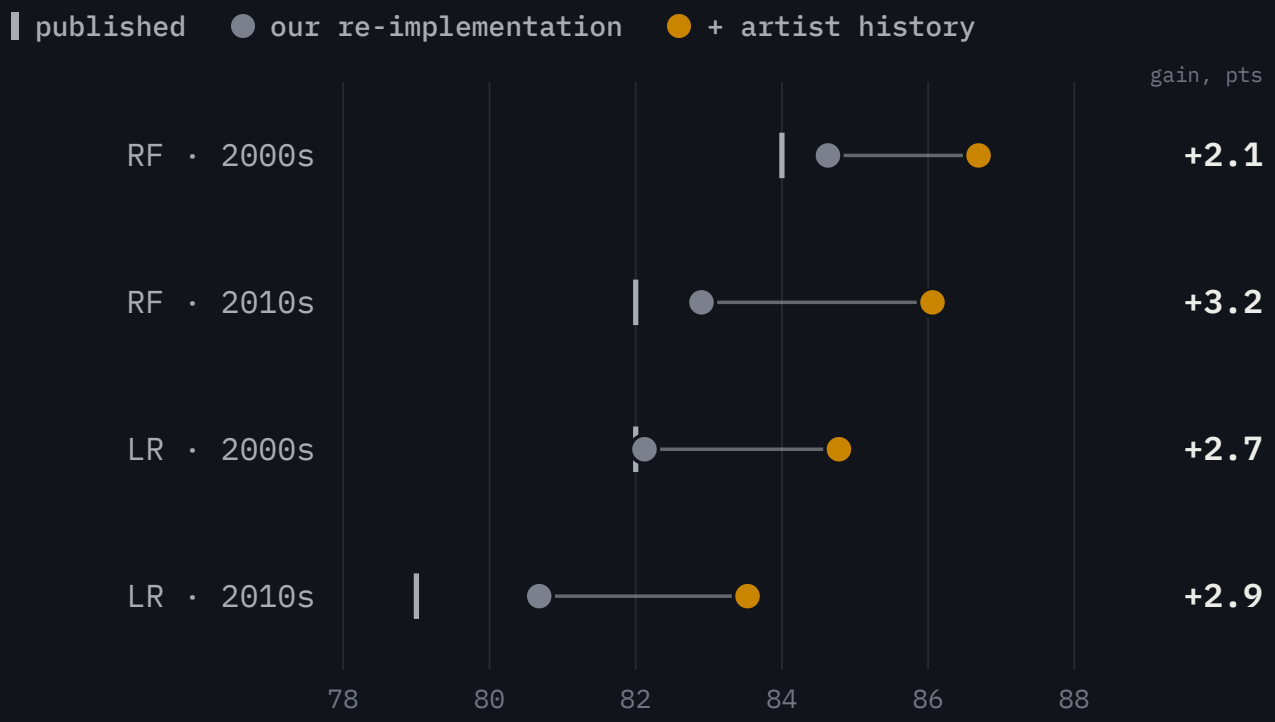


Random Forest, trained on all earlier decades · McNemar p between 10^{-7} and 10^{-21}

Artist history still adds **+1.9 to +3.4 points** on unseen decades, with similar gains for XGBoost.

The decade cannot be used here: the test decade never appears in training.

Their exact models and inputs, plus our feature: better on 50 out of 50 splits



Mean over 50 random 70/30 splits · paper Table V, Fig. 2

One model per decade, 13 audio + chorus_hit + sections — exactly as in the thesis. Our re-implementation lands **on or just above** their published numbers, which validates the protocol.

Adding artist history: **+2.1 to +3.2 points** on every split; the improved models beat the published accuracy on **100 %** of splits.

The Kaggle dataset chose its non-hits partly by artist

DATASET README — A “FLOP” MUST SATISFY ALL OF

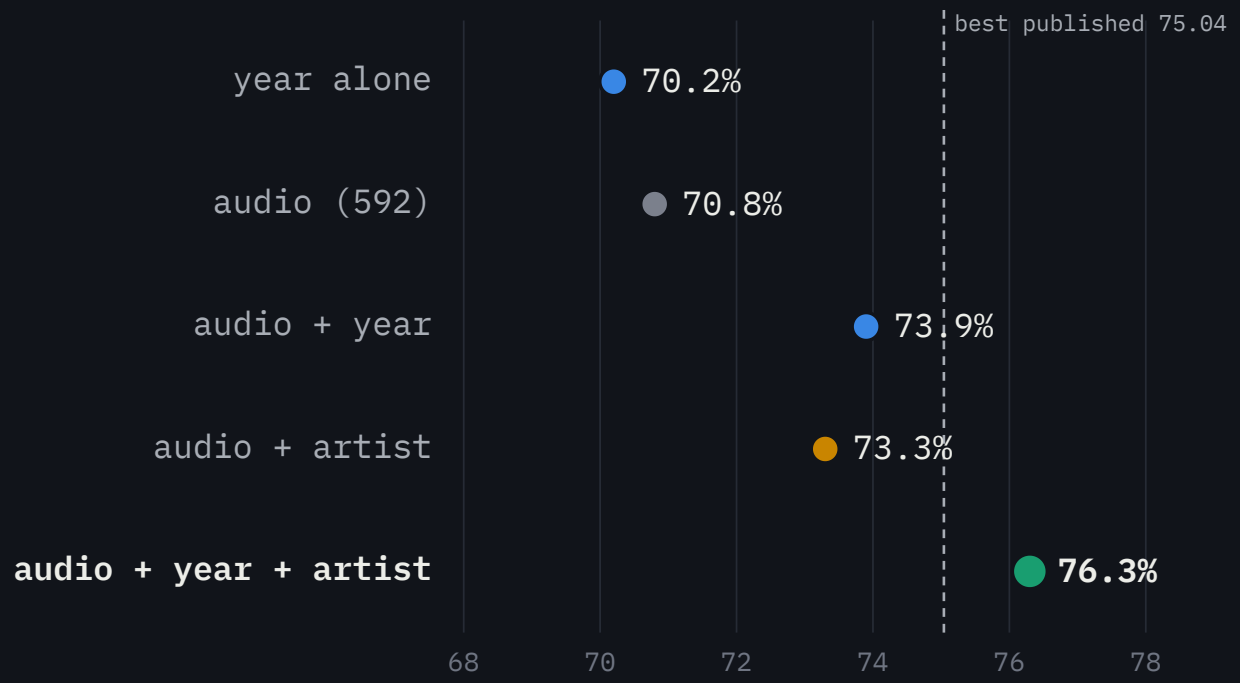
1. The track is not in that decade’s hit list.
2. **The track’s artist is not in that decade’s hit list.**
3. It belongs to a “non-mainstream and/or avant-garde” genre.
4. That genre has no song in the hit list.

96.6%

of songs by an artist with an earlier hit are hits. On this benchmark, “known artist” almost means “hit”.

So the **+5.0** is a **benchmark-specific upper bound**. We need data whose negatives are not chosen by artist → ISMIR 2019.

With artist history we reach 76.3 %, above the best published 75.04 %

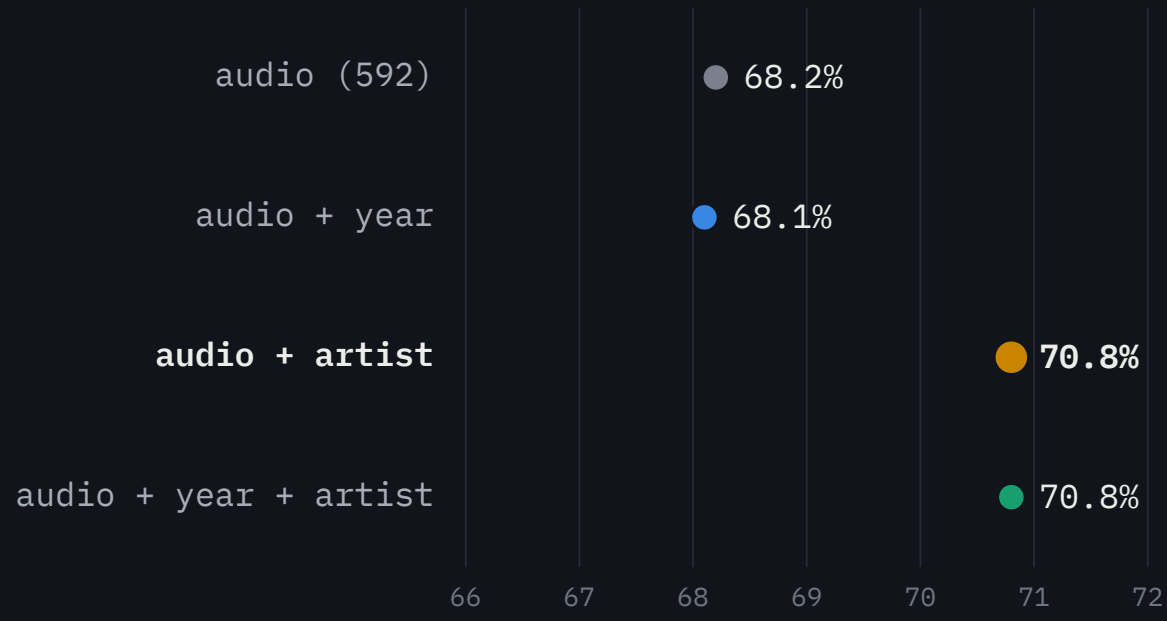


Random Forest on 592 Essentia descriptors · 5×2 CV mean · paper Table VI

Audio + year mirrors the inputs of the best published model and gives 73.9 %. Artist history adds **+2.4 points** (10/10 folds, $p = 7{\cdot}10^{-8}$).

But look at **year alone: 70.2 %** — as good as 592 audio descriptors. Hits in this dataset are older (median 1987) than non-hits (2001).

Match the years: release year becomes useless, artist history still adds +2.6



Same data, one non-hit from the same year for every hit · 5×2 CV · paper Table VI

+2.6

points, p = $3{\cdot}10^{-5}$, 10/10 folds. Negatives chosen **neither by artist nor by date**.

This is the number we defend as the transferable value of artist history. Year: p = .66; year alone falls to 44.8 %, below chance.

EXTERNAL VALIDATION · SPOTGENTRACK, 101,909 TRACKS

With 81 % non-hits, accuracy hides the signal — ROC-AUC shows it



Artist history: AUC **.690** → **.762** (+ decade .777). The 207 low-level audio features add nothing on top of it. 5-fold CV · paper Table VII

Does the model just recognise artists? No — the gains survive when test artists are never seen

HIT PREDICTOR · decade + artist

+5.0 → +5.0

82.2 % grouped (83.2 % random). 25/25 folds, $p = 10^{-13}$.

ISMIR B · artist history

+2.6 → +2.9

10/10 folds, $p = 9{\cdot}10^{-7}$. Year still useless.

ISMIR A · audio + year + artist

75.9%

CI [75.2, 76.6] — still entirely above the published 75.04 %.

All accuracies drop by up to one point — the audio carried a little artist identity — but the gains remain (ISMIR A loses at most half a point). Paper Table VIII

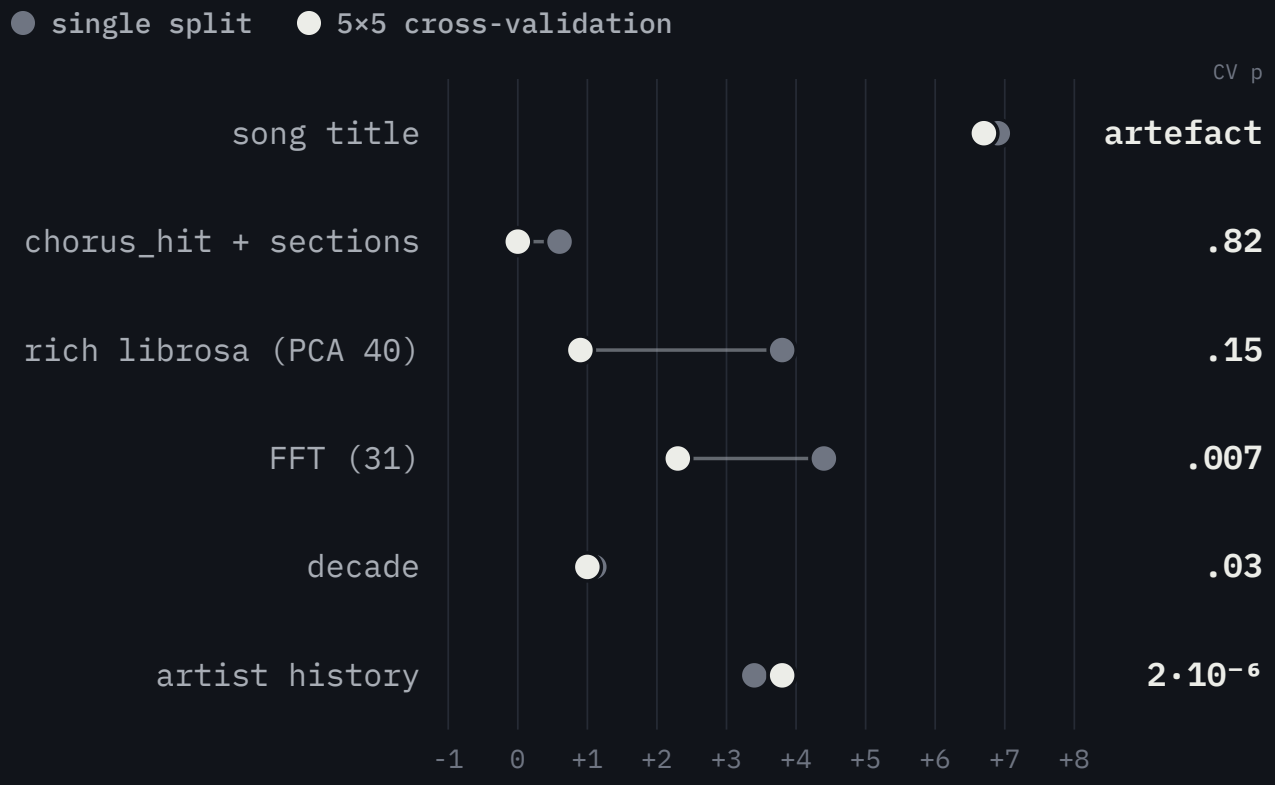
ROBUSTNESS · COLD START

For a new artist, it is the era that helps — not the artist feature

DATASET	HITS NEW / KNOWN	NEW · + ERA	NEW · + ARTIST	KNOWN · + ARTIST
Hit Predictor	39 % / 97 %	+2.2 $p=2\cdot10^{-9}$	+0.4 n.s.	+15.0 $p=3\cdot10^{-14}$
ISMIR A	36 % / 75 %	+3.4 $p=6\cdot10^{-5}$	+0.8 n.s.	+4.4 $p=.003$
ISMIR B (year-matched)	37 % / 72 %	+0.1 n.s.	+1.4 $p=.002$	+5.3 $p=8\cdot10^{-5}$

Gain in points over audio, artist-grouped CV. On the benchmark, artist history's whole gain comes from the 17.6 % of songs by established artists.
Paper Table VIII

A big, significant gain can still be wrong: artefact or luck



Gain over audio in points · top: full benchmark · bottom: 3,562-song audio subset · paper Table IX

ARTEFACT **Song titles +6.7**, more than any musical feature. Titles don't make hits — they reveal that non-hits come from niche genres and catalogue releases.

LUCK **Rich librosa audio**: +3.8 (p = .002) on one split, **+0.9, not significant** in CV. Same for "PCA 40 is the peak" and the family knock-outs.

chorus_hit + sections, often suspected of leakage, add nothing.

With many tests, some pass by chance — so we apply Bonferroni: $p < 0.05 / 60 \approx 8 \cdot 10^{-4}$

SURVIVES — OUR CONCLUSIONS REST ON THESE

- Main gains on the benchmark ($p = 7 \cdot 10^{-17}$)
- All five classifiers
- Head-to-head with Du, all four models
- Temporal hold-out on unseen decades
- Artist history on ISMIR, protocols A and B
- Artist-grouped gains

INDICATIVE ONLY — $8 \cdot 10^{-4} < p < 0.05$

- Decade and FFT gains on the small audio subset
- Most single-split gains on that subset
- SpotGenTrack accuracy, artist alone ($p = 10^{-3}$)
- ISMIR cold-start gains ($p = .002, .003$)

HONESTY · WHAT CHANGED SINCE OUR FIRST PRESENTATION

Six claims from our earlier slides did not survive re-checking

EARLIER SLIDE SAID	WHAT IS ACTUALLY TRUE
Reference paper at 0.877	Middlebrook & Sheik — a **different dataset**, not comparable
Rich audio +3.8, “PCA 40 is the peak”	**+0.9, not significant** in CV; all PCA sizes within noise
Billboard 2000–23 validation: no gain	**Invalid**: 2,911 of 3,397 rows had no audio features
SpotGenTrack “our 2 features alone” .8222	That score was **artist + low-level audio**
Tilburg .878 / .890, “a close comparison”	Exact protocol, 50 splits: **.867 / .861** vs published .84 / .82
Artist history “works on new songs too”	It does **not help new artists** on the benchmark (+0.4, n.s.)

WHAT IS NEW

Artist history is not a new idea. Proving what it is worth is.

ALREADY EXISTED

Middlebrook & Sheik (2019) list an artist past-performance field among chart-derived fields — but their final inputs are not stated and its contribution is never isolated or tested.

OUR CONTRIBUTION

- A **leakage-free** version (strictly earlier periods)
- A significance-tested **head-to-head** on the same data as prior work
- **External validation** with a year-matched control
- **Unseen-artist** and **cold-start** analysis
- An **audit** of artefacts and single-split luck

What this study cannot say

LIMITATIONS

- Artist names match for only 52 % of benchmark songs
- Artist history partly measures **cumulative advantage**, not a song's appeal
- ISMIR: a Random Forest and a different draw of negatives, not their network
- Du: re-drawn splits, since the original R seed cannot be replayed
- Audio clips come from Deezer, not from Spotify

NEXT

- Canonical artist IDs (MusicBrainz / Spotify) to raise the match rate
- Features that help **first-time artists**: label, collaborators, social media
- Rolling-origin evaluation across all decades
- Lyrics and learned audio embeddings, under the same audit
- A benchmark whose negatives ignore artist, genre and date

TAKE-HOME

Knowing which gains are real matters more than the headline number

+5.0

points on the Kaggle benchmark, every classifier, every decade — an upper bound inflated by how non-hits were chosen.

+2.6

points on independent data, years matched, artists unseen: the value we defend.

76.3%

above the best published 75.04 % on ISMIR 2019, and ahead of the only prior model on the benchmark on 50/50 splits.

Song titles, release dates and rich audio on one split looked better still — and were artefacts or chance. Every number: `repro/ scripts → paper_results/`